Riccardo Rettore

Robotics & Physical AI engineer passionate about humanoids, embodied intelligence, and robot autonomy
riccardorettore@gmail.com | riccardo-rettore | RiccardoRettore | portfolio
+39-3407068944 ◦ Trebaseleghe, Padova, IT

Experience

- **Humanoid Robotics & Autonomous Systems Engineer @ EssilorLuxottica** — *May 2026 – Present*
  Quality Control Robotics Intern, Industrial Humanoid Deployment & Autonomous Test Automation — Agordo, Italy
  - Deploying and evaluating **humanoid robot platform** in a real industrial manufacturing environment, developing pipelines for autonomous execution of quality-control tasks previously performed by human operators.
  - Designing perception and decision-making modules that enable robots to interpret multi-modal sensor streams and autonomously adapt task execution to part variation and environmental uncertainty.
  - Designing the automation architecture for **autonomous quality test pipelines** with real-time feedback for closed-loop verification.
- **Co-Founder & CTO @ Resit** — *Dec 2025 - May 2026*
  Smart Seating for healthier employees - resitchair.com — Padova - San Francisco
  - Leading the development of learning-based posture classification models, transforming multi-modal sensor data into real-time user state estimation and ergonomic insights.
  - Driving system architecture and integration across embedded sensing, ML inference, and user-facing applications, enabling closed-loop intelligence from physical interaction to real-time feedback.
- **CDEI Group @ University of Catalunya** — *Feb 2025 – Aug 2025*
  Prof. A. Gracia (UPC), Prof. G. Susto (UniPd) — Barcelona, Spain
  - Developed reinforcement learning–based navigation and control frameworks for autonomous robots using custom **Gymnasium** environments, implementing **PPO** and **SAC** in discrete and continuous action spaces with **Python** and **PyTorch**.
  - Integrated subsurface perception from Ground Penetrating Radar with surface sensing into decision-making pipelines, validating navigation policies in **ROS2**-based simulations with **Gazebo** and visualizing results in **RViz**.
- **Freelancer for Startups** — *Jan 2020 - Jan 2024*
  Full-Stack Developer — Padova, Italy
  - Collaborated with startups to build cross-platform applications using **Flutter** and **Google technologies**.
  - Built both front-end interfaces and back-end systems to deliver seamless, user-friendly solutions.

Education

- **MSc in Control Systems Engineering** — *Oct 2023 - Oct 2025*
  Università degli Studi di Padova, Final Grade: 110/110 — Padova, Italy
  - Final Thesis: Leveraging Ground Penetrating Radar with Deep Reinforcement Learning for Subsurface-Informed Navigation in Autonomous Agricultural Robotics.
  - Relevant Coursework: Reinforcement Learning, Advanced Control Theory, Controls for Robotics Systems, Deep Learning, Machine Learning, Computer Vision, Autonomous Robotics.
  - Erasmus Exchange Program @ Universitat Politècnica de Catalunya, Barcelona, Spain: focused on Reinforcement Learning and Robotics, Physical AI and Autonomous Systems.
- **BSc in Mechatronics Engineering** — *Oct 2020 - July 2023*
  Università degli Studi di Padova, Final Grade: 109/110 — Padova, Italy
  - Final Thesis: Integration between ROS2 and Unity: performance analysis.
  - Relevant Coursework: Applied Mechanics, Analog Electronics, Electrical Machines and Drives, Instrumentation and Measurement, PLCs and Industrial Networks.

Academic Experience

- **Aurora: Object Classification with Oriented Boosting** *July 2024 - Nov 2024*
  Neural Networks and Deep Learning project Padova, Italy
  - Built a 3D object classification pipeline with multi-task CNNs and oriented boosting, including voxelization, augmentation, labeling, and HDF5 preprocessing using **PyTorch**, **NumPy**, and **OpenCV**.
  - Optimized GPU training and evaluated on ModelNet10/40, executing 3D data processing, CNN optimization, and computer vision pipelines.
- **Advanced Control Systems for Double-Wheel Robot** *April 2024 - June 2024*
  Control Engineering Laboratory Project Padova, Italy
  - Designed nominal and robust LQR controllers for a self-balancing two-wheeled robot, modeling dynamics via Lagrange formalism, linearizing around equilibrium, and implementing sensor fusion with complementary filtering in **MATLAB/Simulink**.
  - Performed simulation and on-hardware testing of designed controllers, analyzing system stability, state-feedback, observer performance, and system robustness.

Technical Skills

- **Robotics & Simulation:** ROS2, Gazebo, RViz, MuJoCo, Isaac Gym / Isaac Lab, ROS2, OpenCV
- **ML / RL Frameworks:** PyTorch, TensorFlow, Stable-Baselines3, Gymnasium, Imitation Learning, Transformers
- **Physical AI & Robot Learning:** Large Language models (LLMs), Vision–Language Models (VLMs), Vision–Language–Action (VLA) models
- **Programming:** Python, C++, RAPID, MATLAB/Simulink, Bash/Linux, Git
- **Languages:** Italian (native), English (C1), Spanish (B1)